

Data Analyst Interview Bible (UK 2026)

UK Interview Edition

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2. Data Analyst Interview Bible (UK 2026)

2.1 Part 1 — HR Interview Mastery

First Edition

Authors

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2.2 Part 2 — SQL Bible

This section prepares you for SQL interviews in UK companies using one consistent business dataset:

Evenec Retail.

Topics include:

- SELECT & WHERE
- GROUP BY & HAVING
- JOINS
- CTEs
- Window Functions
- Business Cases
- Full Mock Interview

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5. HR Interview Mastery

6. Questions 1–15

This chapter covers the first fifteen HR interview questions most commonly asked during Junior Data Analyst interviews in the United Kingdom.

The answers are written in **natural B2 English** and are designed to sound confident, professional and conversational.

7. How to Use This Chapter

Each question follows the same structure.

- **Question** – what the interviewer asks.
 - **Recruiter Psychology** – what they are really evaluating.
 - **Model Answer** – natural B2 English.
 - **Bogdan Note** – how to adapt it to your real experience.
 - **Memory Hook** – a quick phrase for remembering the structure.
 - **Follow-up Questions** – what usually comes next.
 - **Common Mistake** – what to avoid.
-

8. Question 1 — Tell me about yourself

This is almost always the first question.

The interviewer is not asking for your life story.

They are checking whether you can introduce yourself clearly.

8.1 Recruiter Psychology

They evaluate:

- structured thinking;
- communication;
- confidence;
- career direction.

8.2 Model Answer

I'm currently working in a dental laboratory in London as a 3D Designer, where I work with digital workflows and attention to detail every day.

Before this role, I ran my own business in Ukraine, where I regularly worked with sales reports and customer information to make business decisions.

That experience made me interested in data analysis, so I completed the GoIT Data Analytics programme and learned SQL, Excel, Power BI and Python.

Now I'm looking for my first Junior Data Analyst role where I can combine business understanding with analytical skills.

Bogdan Note

Use your real transition story.

Business → Dental Lab → GoIT → Data Analyst.

This makes your career change feel intentional.

Memory Hook

Past → Present → Future

8.2.1 Follow-up Questions

- Why did you change careers?
- What did you enjoy about your business?
- Why data analytics?

8.2.2 Common Mistake

Don't say:

"I love numbers."

Instead, connect your answer to real experience.

9. Question 2 — Why do you want to become a Data Analyst?

9.1 Recruiter Psychology

The interviewer wants to know whether your motivation is sustainable.

They're checking if this is a thoughtful career move rather than a temporary idea.

9.2 Model Answer

I enjoy solving business problems with data.

In my previous business I often used sales reports and customer information to make decisions.

I realised I wanted stronger technical skills, so I completed a Data Analytics programme.

Now I want to use both my business experience and analytical skills in a professional team.

💡 Bogdan Note

Mention business decisions instead of saying you've always loved data.

i Memory Hook

Business → Learning → Career

9.2.1 Common Mistake

Don't say:

"Because analysts earn good salaries."

10. Question 3 — Why are you leaving your current job?

10.1 Recruiter Psychology

The interviewer wants to see whether you leave professionally.
They're checking for maturity rather than complaints.

10.2 Model Answer

I've learned valuable skills in my current role, especially attention to detail and digital workflows.
However, my long-term goal is to build my career in data analytics.
I'm looking for a role where I can grow professionally and use the skills I've been developing through my training.

i Memory Hook

Positive → Growth → Future

10.2.1 Follow-up Questions

- What did you learn?
- What will you miss?

10.2.2 Common Mistake

Never criticise your current employer.

11. Practice Break

Answer these aloud.

1. Tell me about yourself.
2. Why Data Analytics?
3. Why are you leaving your current job?

Target time:

45–60 seconds each.

12. Question 4 — Why do you want to work for our company?

12.1 Recruiter Psychology

This question checks whether you prepared.

A candidate who researches the company usually appears more motivated.

12.2 Model Answer

I like that your company makes decisions based on data and invests in digital improvement.

I also appreciate the opportunity to learn from experienced analysts and work on real business problems.

I'm looking for a place where I can grow while contributing from the beginning.

Bogdan Note

Before every interview, spend five minutes checking the company's website, LinkedIn page and recent news.

12.2.1 Company Research Checklist

- Website
- LinkedIn
- News
- Products
- Values

12.2.2 Common Mistake

Never answer:

"I just need a job."

13. Question 5 — What are your strengths?

13.1 Recruiter Psychology

The interviewer wants strengths supported by evidence.

13.2 Model Answer

I'd highlight three strengths.

First, attention to detail.

Second, problem-solving.

Third, business understanding from running my own company.

 Bogdan Note

Always attach a real example to each strength.

13.2.1 Memory Hook

Three strengths.

13.2.2 Common Mistake

Listing ten strengths without examples.

14. Question 6 — What is your biggest weakness?

14.1 Recruiter Psychology

This isn't a trap.

They want to see self-awareness.

14.2 Model Answer

Earlier I sometimes spent too much time trying to make every detail perfect.

I've learned to balance quality with deadlines by prioritising the most important tasks first.

14.2.1 Good Weaknesses

- perfectionism
- asking for help too late
- public speaking while improving

14.2.2 Common Mistake

Don't use fake weaknesses like:

"I work too hard."

15. Practice Break

Ask yourself:

- Which strength has the strongest real example?
 - Which weakness shows genuine improvement?
-

16. Question 7 — Tell me about a challenge you faced.

16.1 Recruiter Psychology

This question measures problem-solving.
Use STAR.

16.2 Model Answer

16.2.1 Situation

Our workflow had delays.

16.2.2 Task

Reduce processing time.

16.2.3 Action

Reviewed the workflow and improved communication.

16.2.4 Result

The process became smoother and deadlines became easier to meet.

i Memory Hook

STAR
Situation → Task → Action → Result

16.2.5 Bogdan Note

Later we'll replace this with your strongest real business example.

17. Question 8 — Tell me about a mistake you made.

17.1 Recruiter Psychology

Recruiters don't expect perfection.
They expect accountability.

17.2 Model Answer

In one project I focused too much on one part of the work before checking the wider process.
I realised this created unnecessary delays.
Since then I've started reviewing priorities earlier and communicating progress more regularly.

17.2.1 Common Mistake

Never blame other people. _____

18. Question 9 — How do you prioritise your work?

18.1 Recruiter Psychology

Analysts constantly balance multiple requests.

The interviewer wants to understand your decision process.

18.2 Model Answer

I usually start by identifying deadlines and business impact.

Then I break larger tasks into smaller steps and review priorities during the day if something changes.

i Memory Hook

Impact → Deadline → Dependencies

18.2.1 Common Mistake

Saying:

“I just do everything.”

19. Practice Break

Answer Questions 7–9 in under **90 seconds** each.
Focus on speaking naturally rather than perfectly.

20. Question 10 — Tell me about working with difficult people.

20.1 Recruiter Psychology

This question tests emotional intelligence.

20.2 Model Answer

I try to understand their perspective first.

Then I focus on solving the problem instead of making it personal.

Clear communication usually helps avoid misunderstandings.

20.2.1 Bogdan Note

Use a real workplace example without criticising colleagues.

21. Question 11 — Describe teamwork.

21.1 Recruiter Psychology

Junior analysts rarely work alone.
They're checking collaboration.

21.2 Model Answer

I enjoy working in teams because different people bring different strengths.
In my current role I regularly communicate with colleagues to make sure work moves smoothly between different stages.

21.2.1 Memory Hook

Communication → Reliability → Support.

22. Question 12 — How do you handle pressure?

22.1 Recruiter Psychology

The interviewer wants confidence under deadlines.

22.2 Model Answer

I stay calm by breaking large tasks into smaller actions.

I also keep track of priorities so I focus on the most important work first.

i Memory Hook

Break → Prioritise → Execute

22.2.1 Common Mistake

Don't pretend pressure never affects you.

23. Question 13 — Where do you see yourself in five years?

23.1 Recruiter Psychology

They're checking ambition without unrealistic expectations.

23.2 Model Answer

In five years I'd like to become a confident Data Analyst with strong technical skills and experience solving real business problems.

I'd also like to mentor newer colleagues as I continue developing professionally.

23.2.1 Common Mistake

Don't say:

"I want your manager's job."

24. Question 14 — Why should we hire you?

24.1 Recruiter Psychology

This is your value statement.

24.2 Model Answer

I bring a combination of business experience, attention to detail and technical skills.

I've already invested time in learning SQL, Excel, Power BI and Python, and I'm motivated to continue growing while contributing to the team.

💡 Bogdan Note

Keep this answer around 30 seconds.

24.2.1 Memory Hook

Business + Technical + Motivation.

25. Question 15 — Do you have any questions for us?

25.1 Recruiter Psychology

This is often the final impression.
Good questions show curiosity.

25.1.1 Model Answer

I'd love to know what success looks like during the first three months in this role.
I'd also be interested in learning about the team's projects and development opportunities.

25.1.2 Five Excellent Questions

- What does success look like in the first three months?
- What projects would I work on first?
- How does the team collaborate?
- What learning opportunities do junior analysts receive?
- Which tools does the team use most often?

25.1.3 Common Mistake

Never answer:

"No, I don't have any questions."

26. Chapter Summary

26.1 Top Five Memory Hooks

Concept	Hook
Self-introduction	Past → Present → Future
STAR	Situation → Task → Action → Result
Prioritisation	Impact → Deadline → Dependencies
Pressure	Break → Prioritise → Execute
Company Research	Website → LinkedIn → News

27. 5-Minute Interview Drill

Set a timer.

Answer these five questions without reading your notes.

Question	Target
Tell me about yourself	60 s
Why Data Analytics?	45 s
Biggest strength	30 s
Biggest weakness	45 s
Why this company?	45 s

Listen to your recording afterwards.

Improve **one sentence**, not the whole answer.

Small improvements repeated consistently create confident interviews.

28. HR Interview Mastery

29. Questions 16–30

This chapter continues the HR interview preparation with behavioural and workplace questions that frequently appear during Junior Data Analyst interviews in the UK.

The goal is not to memorise perfect sentences but to understand how recruiters evaluate your thinking and communication.

30. Question 16 — Tell me about a time you learned something new quickly.

30.1 Recruiter Psychology

Recruiters want evidence that you can learn new tools and adapt to change.

30.2 Model Answer

When I decided to move into data analytics, I needed to learn several new tools in a relatively short time.

I completed the GoIT Data Analytics programme, practised SQL, Excel, Power BI and Python, and built projects to apply what I learned.

That experience taught me how to organise my learning and stay consistent.

 Bogdan Note

This is one of your strongest career-transition stories.

 Memory Hook

Learn → Practice → Apply

30.2.1 Common Mistake

Don't simply say:

"I'm a fast learner."

Show evidence.

31. Question 17 — Describe a time you improved a process.

31.1 Recruiter Psychology

Analysts improve processes constantly.

Recruiters want initiative.

31.2 Model Answer

In my previous work I reviewed how tasks moved through our workflow.

I noticed unnecessary delays between steps, suggested a clearer process and improved communication.

The workflow became more organised and easier to manage.

31.2.1 Memory Hook

Observe → Improve → Result.

32. Question 18 — Tell me about a time you worked with data.

32.1 Recruiter Psychology

Even if this isn't a formal analyst role, recruiters want transferable experience.

32.2 Model Answer

In my business I regularly reviewed sales information and customer data to understand performance.
I used that information when making decisions about products and priorities.
That experience became one of the reasons I became interested in data analytics.

 Bogdan Note

This answer connects your business experience directly to analytics.

33. Practice Break

Answer Questions 16–18 without reading.

Target:

- 60 seconds each.

34. Question 19 — How do you deal with repetitive tasks?

34.1 Recruiter Psychology

Data work often includes repetitive processes.
The interviewer wants discipline.

34.2 Model Answer

I try to stay focused by understanding why the task matters.
I also look for opportunities to organise my work more efficiently while maintaining accuracy.

34.2.1 Memory Hook

Purpose → Accuracy → Efficiency.

34.2.2 Common Mistake

Don't say repetitive work is boring.

35. Question 20 — Tell me about receiving feedback.

35.1 Recruiter Psychology

Can you accept feedback professionally?

35.2 Model Answer

I see feedback as an opportunity to improve.

In previous roles I've received suggestions that helped me become more organised and communicate more clearly.

I try to apply feedback as quickly as possible.

35.2.1 Memory Hook

Listen → Apply → Improve.

36. Question 21 — Tell me about giving feedback.

36.1 Recruiter Psychology

This measures communication and professionalism.

36.2 Model Answer

I try to give feedback respectfully and focus on the work rather than the person.
I usually explain what worked well first before suggesting improvements.

36.2.1 Common Mistake

Avoid sounding confrontational.

37. Question 22 — How do you organise your work?

37.1 Recruiter Psychology

Organisation is essential for analysts.

37.2 Model Answer

I usually start by identifying priorities and deadlines.

Then I create a simple task list and review my progress during the day.

This helps me stay organised without losing focus.

i Memory Hook

Plan → Track → Review.

38. Practice Break

Ask yourself:

- Which answer sounds the most natural?
 - Which example feels strongest?
-

39. Question 23 — Tell me about a time you solved a problem.

39.1 Recruiter Psychology

Problem-solving is one of the core skills for analysts.

39.2 Model Answer

I once noticed that a workflow was creating unnecessary delays.

I reviewed the process, identified where communication was slowing things down and suggested a clearer approach.

The workflow became more efficient afterwards.

39.2.1 Memory Hook

Problem → Analysis → Solution → Result.

40. Question 24 — How do you stay motivated?

40.1 Recruiter Psychology

Recruiters want consistent motivation.

40.2 Model Answer

I stay motivated when I can see progress.

I enjoy learning new skills and solving practical problems, so I like setting small goals and improving step by step.

40.2.1 Common Mistake

Don't rely only on external motivation.

41. Question 25 — What does success mean to you?

41.1 Recruiter Psychology

The interviewer wants realistic professional values.

41.2 Model Answer

For me, success means delivering reliable work, continuing to learn and helping the team achieve good results.

I also think success includes building skills that allow me to take on greater responsibility over time.

41.2.1 Memory Hook

Reliable → Learn → Grow.

42. Practice Break

Record your answers to Questions 23–25.

Listen for:

- clarity
 - pace
 - confidence
-

43. Question 26 — Tell me about working under a deadline.

43.1 Recruiter Psychology

Deadlines are part of everyday analytical work.

43.2 Model Answer

When I work under deadlines, I first identify the highest-priority tasks.
Then I break the work into manageable steps and monitor my progress.
This helps me stay calm and deliver quality work.

43.2.1 Memory Hook

Prioritise → Break Down → Deliver.

44. Question 27 — Describe a situation where you took initiative.

44.1 Recruiter Psychology

Recruiters want proactive people.

44.2 Model Answer

I noticed an opportunity to improve part of our workflow without being asked.

I suggested a practical change, discussed it with the team and helped implement it.

The process became smoother afterwards.

 Bogdan Note

Use examples where you solved problems instead of waiting for instructions.

45. Question 28 — How do you deal with uncertainty?

45.1 Recruiter Psychology

Analysts often work with incomplete information.

45.2 Model Answer

I try to gather as much relevant information as possible before making a decision.

If something is unclear, I ask questions instead of making assumptions.

I believe good communication reduces uncertainty.

45.2.1 Memory Hook

Gather → Ask → Decide.

46. Question 29 — Describe a time you had multiple priorities.

46.1 Recruiter Psychology

This tests workload management.

46.2 Model Answer

I've had situations where several important tasks needed attention at the same time.

I reviewed deadlines, considered business impact and completed the highest-priority work first.

That helped me meet expectations without losing quality.

46.2.1 Common Mistake

Don't say you simply worked longer hours.

47. Question 30 — Why should we believe you'll succeed as a Junior Data Analyst without previous analyst experience?

47.1 Recruiter Psychology

This is one of the most important transition questions.
Recruiters want confidence supported by evidence.

47.2 Model Answer

Although this would be my first professional Data Analyst role, I already have experience making business decisions using information, working carefully with digital workflows and learning new technical skills.

I've invested significant time in SQL, Excel, Power BI and Python, and I'm ready to continue learning while contributing to the team from day one.

Bogdan Note

This is your strongest closing answer for career-transition interviews.

Memory Hook

Experience + Skills + Motivation.

47.2.1 Common Mistake

Never apologise for being junior.
Instead, emphasise what you already bring.

48. Chapter Summary

48.1 Five Key Memory Hooks

Concept	Memory Hook
Learning	Learn → Practice → Apply
Organisation	Plan → Track → Review
Problem Solving	Problem → Analysis → Solution
Deadlines	Prioritise → Break Down → Deliver
Career Transition	Experience + Skills + Motivation

49. 5-Minute Interview Drill

Answer these questions without looking at your notes.

Question	Target
Learned something quickly	60 s
Improved a process	60 s
Worked with data	45 s
Solved a problem	60 s
Why you'll succeed as a Junior Data Analyst	60 s

Focus on speaking naturally rather than memorising every word.

Your goal is confidence, not perfection.

50. HR Interview Mastery

51. Questions 31–40

This chapter focuses on ownership, communication, adaptability and professional judgement.

These questions are common in behavioural interviews because recruiters want to understand how you think when situations become less predictable.

52. Question 31 — Tell me about a time you had to learn from failure.

52.1 Recruiter Psychology

The interviewer wants to see resilience.

They're looking for someone who reflects, learns and improves instead of becoming defensive.

52.2 Model Answer

One experience taught me that focusing only on immediate tasks without reviewing the bigger picture can create unnecessary delays.

I reflected on what happened, changed how I planned my work and started checking priorities earlier.

Since then I've become more organised when managing multiple responsibilities.

Bogdan Note

Focus on the lesson rather than the failure itself.

Memory Hook

Reflect → Adjust → Improve.

52.2.1 Common Mistake

Don't choose a failure that makes you sound careless or irresponsible.

53. Question 32 — Describe a time you had to explain something complicated.

53.1 Recruiter Psychology

Analysts often explain technical information to non-technical people.
Communication matters as much as technical knowledge.

53.2 Model Answer

I try to explain complex ideas using simple language and practical examples.
I usually avoid unnecessary technical terms and check whether the other person understands before continuing.
Clear communication helps everyone make better decisions.

53.2.1 Memory Hook

Simple → Example → Confirm.

53.2.2 Follow-up Questions

- How do you know someone understands?
 - Can you give an example?
-

54. Question 33 — How do you handle changing priorities?

54.1 Recruiter Psychology

Business priorities change constantly.
They're checking flexibility.

54.2 Model Answer

When priorities change, I first understand what has become most important for the business.
Then I reorganise my tasks and communicate if timelines need to change.
Staying flexible helps me deliver the most valuable work first.

i Memory Hook

Understand → Reorganise → Communicate.

54.2.1 Common Mistake

Don't say changing priorities frustrates you.

55. Practice Break

Answer Questions 31–33.

Focus on speaking naturally.

Target:

- 60 seconds each.
-

56. Question 34 — Tell me about a time you had to make a difficult decision.

56.1 Recruiter Psychology

They're evaluating judgement.

56.2 Model Answer

I've had situations where I needed to balance several competing priorities.

I gathered the available information, considered the potential impact and made the decision that best supported the overall goal.

Afterwards I reviewed the outcome so I could learn from the experience.

56.2.1 Memory Hook

Gather → Decide → Review.

57. Question 35 — What would you do if you didn't know the answer?

57.1 Recruiter Psychology

This is an honesty test.

Good analysts don't pretend to know everything.

57.2 Model Answer

If I didn't know the answer, I'd first research the problem using available resources.

If I still needed help, I'd ask a colleague while explaining what I'd already tried.

I believe learning efficiently is more important than pretending to know everything.

 Bogdan Note

This answer shows independence before asking for help.

57.2.1 Common Mistake

Never say you'd guess. _____

58. Question 36 — Tell me about a time you worked independently.

58.1 Recruiter Psychology

Junior analysts need supervision, but they also need initiative.

58.2 Model Answer

In previous roles I've managed responsibilities independently by planning my work and tracking my progress.

When necessary, I communicated updates and asked questions early rather than waiting until problems became bigger.

That helped me stay responsible while working efficiently.

58.2.1 Memory Hook

Plan → Execute → Update.

59. Practice Break

Record yourself answering Questions 34–36.

Listen for:

- confidence
 - clarity
 - pace
-

60. Question 37 — Describe a situation where you had to be very accurate.

60.1 Recruiter Psychology

Accuracy is one of the most important qualities for analysts.

60.2 Model Answer

In my current work, accuracy is essential because small mistakes can affect later stages of the process. I developed habits like double-checking important details and following consistent workflows. Those habits are directly relevant to analytical work.

 Bogdan Note

This is a great opportunity to reference your dental laboratory experience.

 Memory Hook

Check → Verify → Deliver.

60.2.1 Common Mistake

Don't simply say you're detail-oriented.
Explain how you ensure accuracy.

61. Question 38 — How do you react when someone disagrees with you?

61.1 Recruiter Psychology

Conflict management matters.
They're looking for professionalism.

61.2 Model Answer

I try to understand why the other person sees the situation differently.
Then I focus on discussing facts rather than personalities.
Even if we disagree, I believe respectful communication usually leads to better decisions.

61.2.1 Memory Hook

Listen → Understand → Discuss Facts.

62. Question 39 — Tell me about a goal you achieved.

62.1 Recruiter Psychology

They want evidence that you finish what you start.

62.2 Model Answer

One goal I achieved was successfully completing my transition toward data analytics.

I balanced work with learning SQL, Excel, Power BI and Python while completing practical projects.

That experience showed me the importance of consistency over time.

 Bogdan Note

Use your career transition as a personal achievement.

62.2.1 Common Mistake

Choose a goal with a measurable result whenever possible.

63. Question 40 — Why should we choose you over another candidate?

63.1 Recruiter Psychology

This is your positioning statement.

They're asking what makes you different.

63.2 Model Answer

I believe I offer a valuable combination of business experience, technical learning and a strong willingness to grow.

I've already demonstrated that I can learn new skills, adapt to change and work carefully with detailed processes.

I'm ready to bring that mindset into my first Junior Data Analyst role.

i Memory Hook

Business + Learning + Reliability.

63.2.1 Common Mistake

Don't compare yourself negatively with other candidates.

Focus on your own value.

64. Chapter Summary

64.1 Five Key Memory Hooks

Concept	Memory Hook
Learning from failure	Reflect → Adjust → Improve
Explaining complexity	Simple → Example → Confirm
Changing priorities	Understand → Reorganise → Communicate
Decision making	Gather → Decide → Review
Accuracy	Check → Verify → Deliver

65. STAR Practice

Complete these prompts using the STAR framework.

Prompt	Time
Learned from failure	90 s
Explained something complicated	90 s
Worked independently	90 s
Achieved a goal	90 s

Remember:

- Situation
 - Task
 - Action
 - Result
-

66. End-of-Chapter Reflection

Ask yourself these questions.

- Which answer sounds most natural?
- Which story feels strongest?
- Which answer still sounds memorised?
- Which example could become more specific?

Your goal is not perfect English.

Your goal is sounding like a confident future colleague.

67. HR Interview Mastery

68. Questions 41–50

This final chapter completes the HR Interview Mastery section with advanced behavioural questions, salary negotiation strategies for the UK market, recruiter red flags, a realistic mock interview and practical checklists.

The goal is to help you finish an interview with confidence.

69. Question 41 — Tell me about a time you disagreed with a decision.

69.1 Recruiter Psychology

They're checking whether you can disagree professionally.

69.2 Model Answer

I try to understand the reasoning behind a decision before reacting.

If I have concerns, I explain them respectfully and focus on facts rather than emotions.

Even when a different decision is made, I continue supporting the team's goal.

i Memory Hook

Listen → Explain → Support.

69.2.1 Common Mistake

Don't describe yourself as someone who always argues.

70. Question 42 — Tell me about adapting to change.

70.1 Recruiter Psychology

Analysts work in changing environments.

70.2 Model Answer

I've experienced several major changes during my career, including moving to London and changing careers into data analytics.

Those experiences taught me how to learn quickly and stay flexible while continuing to work toward long-term goals.

 Bogdan Note

Your move to London is a strong real-life example of adaptability.

70.2.1 Memory Hook

Adapt → Learn → Grow. _____

71. Question 43 — How do you handle constructive criticism?

71.1 Recruiter Psychology

They're evaluating coachability.

71.2 Model Answer

I appreciate constructive feedback because it helps me improve.

I try to understand the suggestion, apply it in my work and review whether the change produces better results.

71.2.1 Memory Hook

Listen → Apply → Improve.

72. Practice Break

Answer Questions 41–43.

Target:

- 45–60 seconds each.
-

73. Question 44 — Describe a situation where you helped someone.

73.1 Recruiter Psychology

They're checking collaboration.

73.2 Model Answer

I enjoy helping colleagues when I can.

Sometimes sharing knowledge or explaining a process saves time for the whole team.

I believe teamwork becomes stronger when people support each other.

73.2.1 Common Mistake

Don't make yourself sound like the only person who solves problems.

74. Question 45 — How do you stay organised when learning new skills?

74.1 Recruiter Psychology

Junior analysts learn constantly.

74.2 Model Answer

I usually divide large topics into smaller learning goals.

Then I practise regularly and apply what I've learned through real exercises or projects.

That approach helped me learn SQL, Excel, Power BI and Python.

i Memory Hook

Small Goals → Practice → Projects.

75. Question 46 — Tell me about making a decision with limited information.

75.1 Recruiter Psychology

Real business decisions rarely have perfect information.

75.2 Model Answer

When information is limited, I collect the most relevant facts available and avoid making unnecessary assumptions.

If needed, I ask questions to reduce uncertainty before deciding.

My goal is making the most informed decision possible.

75.2.1 Memory Hook

Gather → Clarify → Decide.

76. Practice Break

Ask yourself:

- Which answer feels most natural?
 - Which example sounds strongest?
-

77. Question 47 — What motivates you to keep learning?

77.1 Recruiter Psychology

They want long-term learners.

77.2 Model Answer

I enjoy seeing measurable progress.

Learning new skills becomes more motivating when I can immediately apply them to practical problems.

That's one reason I enjoy data analytics.

77.2.1 Memory Hook

Progress → Application → Motivation.

78. Question 48 — Describe a time you worked carefully with details.

78.1 Recruiter Psychology

Accuracy is critical.

78.2 Model Answer

In my current work, small details can affect later stages of production.
I developed habits like checking important details twice and following consistent workflows.
Those habits are directly relevant to analytical work.

 Bogdan Note

Your dental laboratory experience is one of your strongest proof points.

78.2.1 Memory Hook

Check → Verify → Deliver. _____

79. Question 49 — What would your previous colleagues say about you?

79.1 Recruiter Psychology

They're checking self-awareness.

79.2 Model Answer

I think they'd describe me as reliable, organised and willing to help.

I'd also hope they'd say I'm someone who learns quickly and stays calm when solving problems.

79.2.1 Common Mistake

Don't exaggerate.

Keep it believable.

80. Question 50 — Is there anything else you'd like us to know?

80.1 Recruiter Psychology

This is your final opportunity.
Finish confidently.

80.2 Model Answer

I'd like you to know that I'm genuinely committed to building my career in data analytics.
I've already invested significant time in developing technical skills while gaining valuable experience in business and digital workflows.
I'm excited about contributing to a team where I can continue learning and creating value.

i Memory Hook

Commitment → Skills → Value.

80.2.1 Final Impression Tip

End with confidence.
Pause.
Smile.

81. Chapter Summary

81.1 Five Final Memory Hooks

Concept	Memory Hook
Disagreement	Listen → Explain → Support
Adaptability	Adapt → Learn → Grow
Learning	Small Goals → Practice → Projects
Accuracy	Check → Verify → Deliver
Final Answer	Commitment → Skills → Value

82. Salary Negotiation (UK £35–45k)

For your situation, salary negotiation is especially important because you're targeting London.

82.1 When should you discuss salary?

The safest order is:

1. HR screening.
2. Technical interview.
3. Final discussion.
4. Salary conversation.

Avoid becoming the first person to mention a number unless asked directly.

82.2 If HR asks first

What salary are you expecting?

82.2.1 Best Answer (B2)

Based on my research and the responsibilities of this role, I'm looking for something in the region of **£35,000 to £45,000**, depending on the overall package and growth opportunities.

Why this works:

- sounds flexible;
 - gives a range;
 - avoids sounding demanding.
-

82.3 If they offer £32,000

Don't reject immediately.

82.3.1 Good Response

Thank you for the offer.

I appreciate it.

Based on my research and my skills, I was expecting something closer to **£35,000**.

Is there any flexibility?

Professional.

Calm.

Confident.

82.4 If they ask for your minimum salary

Don't say:

"I'll take anything."

Instead:

I'm looking for a package that reflects the responsibilities of the role, ideally starting around **£35,000**.

82.5 Salary Negotiation Checklist

- Stay calm.
 - Pause before answering.
 - Give a range.
 - Mention market research.
 - Stay professional.
-

83. Recruiter Red Flags

Sometimes interviews reveal problems about the employer.

83.1 Green Flags

- Clear role expectations.
- Friendly communication.
- Learning opportunities.
- Structured interview.
- Honest answers.

83.2 Yellow Flags

- Unclear responsibilities.
- Frequent interruptions.
- Constantly changing answers.

Ask more questions.

83.3 Red Flags

- Nobody can explain the role.
- Unrealistic workload.
- Pressure to accept immediately.
- Negative comments about previous employees.
- No onboarding plan.

Remember:

You're interviewing the company too.

84. Mini Mock Interview

Practice without reading.

84.1 Interview Starts

Interviewer

Tell me about yourself.

(Pause.)

Answer.

Interviewer

Why Data Analytics?

(Pause.)

Answer.

Interviewer

Tell me about a challenge you solved.

(Pause.)

Answer.

Interviewer

Why should we hire you?

(Pause.)

Answer.

Interviewer

Do you have any questions for us?

(Pause.)

Answer.

85. Self-Evaluation

Score yourself.

Skill	Score
Confidence	1–5
Clarity	1–5
Structure	1–5
Natural English	1–5
Examples	1–5

Total:

___/25

Repeat until you consistently score above **20**.

86. Final Interview Checklist

86.1 The Day Before

- Research the company.
- Review your CV.
- Practise five key answers.
- Prepare questions.
- Charge your laptop.
- Check your internet.

86.2 One Hour Before

- Open the meeting link.
- Prepare water.
- Close distractions.
- Keep notes nearby.

86.3 During the Interview

- Smile.
- Speak slowly.
- Pause before answering.
- Use STAR.
- Ask questions.

86.4 After the Interview

Within 24 hours, send a short thank-you message.

Example:

Thank you for your time today.

I enjoyed learning more about the role and the team.

I appreciate the opportunity and look forward to hearing from you.

Professional.

Simple.

Effective.

87. Part 1 Complete

Congratulations.

You have now completed **HR Interview Mastery**.

Before moving to the SQL Bible, make sure you can confidently answer all **50 questions** without memorising every word.

The goal is sounding like a future colleague—not sounding like you’ve memorised a script.

II

181	Running Total	385
182	Moving Average	387
183	Window Function Cheat Sheet	389
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Part II — SQL Bible

88. SQL Bible

89. Sprint 1 — Foundations

89.1 SELECT • WHERE • ORDER BY • LIMIT • DISTINCT

Welcome to the SQL Bible.

This book is designed for **Junior Data Analyst interviews in the UK**.

Instead of learning isolated syntax, you'll solve realistic business problems using one consistent dataset.

90. Introducing the Evenec Retail Dataset

Throughout the SQL Bible we'll work with one fictional company.

Evenec Retail

An online retail business selling products across the UK.

This makes interview practice much closer to real companies like Tesco, Amazon, Deliveroo or Wise.

90.1 Database Structure

Table	Purpose
customers	Customer information
orders	Orders placed
order_items	Products inside orders
products	Product catalogue
payments	Payment information
employees	Internal staff

91. Main Tables

91.1 customers

Column	Type
customer_id	INT
first_name	TEXT
last_name	TEXT
city	TEXT
signup_date	DATE

91.2 orders

Column	Type
order_id	INT
customer_id	INT
order_date	DATE
amount	DECIMAL
status	TEXT

91.3 products

Column	Type
product_id	INT
product_name	TEXT
category	TEXT
price	DECIMAL

92. Memory Map

Throughout this book remember:

Customer places an Order.

Order contains Items.

Items reference Products.

This simple relationship helps you understand joins later.

93. Question 1 — Show Every Customer

93.1 Recruiter Psychology

This tests whether you understand the most fundamental SQL operation.

93.2 Business Task

The manager wants to see every customer.

93.3 SQL Solution

```
SELECT *  
FROM customers;
```

93.4 Explain It (B2)

This query selects every column from the customers table.

93.5 Memory Hook

SELECT = Choose.

93.6 Common Mistake

Using `SELECT ALL`.

`SELECT *` already returns all columns.

93.7 Practice Challenge

Return only:

- first_name
 - city
-

94. Question 2 — Select Specific Columns

94.1 Recruiter Psychology

Interviewers prefer selecting only necessary columns.

94.2 Business Task

Display customer names and cities.

94.3 SQL Solution

```
SELECT first_name,  
       city  
FROM customers;
```

94.4 Explain It (B2)

Instead of returning every column, I only select the information I need.

94.5 Memory Hook

Only what matters.

94.6 Common Mistake

Using `SELECT *` in every query.

95. Question 3 — Filter London Customers

95.1 Recruiter Psychology

This checks your ability to filter data.

95.2 Business Task

Find customers living in London.

95.3 SQL Solution

```
SELECT first_name,  
       city  
FROM customers  
WHERE city = 'London';
```

95.4 Explain It (B2)

First I filter the rows where the city is London, then I return the required columns.

95.5 Memory Hook

WHERE filters rows.

95.6 Common Mistake

Using = incorrectly with text.

Always use quotes.

96. Quick Drill

Without looking:

Write a query that finds customers from Manchester.

97. Question 4 — Orders Above £100

97.1 Recruiter Psychology

Numeric filtering appears constantly during interviews.

97.2 Business Task

Show orders above £100.

97.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
WHERE amount > 100;
```

97.4 Explain It (B2)

I filter orders where the amount is greater than one hundred.

97.5 Memory Hook

Greater than means bigger money.

97.6 Common Mistake

Using `>=` when the question says “above”.

98. Question 5 — Multiple Conditions

98.1 Recruiter Psychology

Interviewers want to see logical operators.

98.2 Business Task

Find completed orders above £200.

98.3 SQL Solution

```
SELECT order_id,  
       amount,  
       status  
FROM orders  
WHERE amount > 200  
       AND status = 'Completed';
```

98.4 Explain It (B2)

Both conditions must be true.

98.5 Memory Hook

AND = Both.
OR = Either.

98.6 Common Mistake

Forgetting quotes around text.

99. Question 6 — OR Condition

99.1 Business Task

Find customers from London or Birmingham.

99.2 SQL Solution

```
SELECT first_name,  
       city  
FROM customers  
WHERE city = 'London'  
       OR city = 'Birmingham';
```

99.3 Explain It (B2)

Either city matches.

99.4 Memory Hook

OR opens another door. _____

100. Interview Tip

Many interviewers ask:

“Can you explain your query?”

Always explain:

1. Filter.
 2. Select.
 3. Result.
-

101. Question 7 — Sort Highest Revenue First

101.1 Recruiter Psychology

Sorting is one of the most common interview tasks.

101.2 Business Task

Show the largest orders first.

101.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
ORDER BY amount DESC;
```

101.4 Explain It (B2)

I sort the amounts from highest to lowest.

101.5 Memory Hook

DESC = Biggest first.

ASC = Smallest first.

101.6 Common Mistake

Forgetting DESC.

Default sorting is ascending.

102. Question 8 — Oldest Customers First

102.1 Business Task

Sort customers by signup date.

102.2 SQL Solution

```
SELECT first_name,  
       signup_date  
FROM customers  
ORDER BY signup_date ASC;
```

102.3 Explain It (B2)

The earliest customers appear first.

103. Question 9 — Top Five Orders

103.1 Recruiter Psychology

LIMIT appears frequently.

103.2 Business Task

Return only the five largest orders.

103.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
ORDER BY amount DESC  
LIMIT 5;
```

103.4 Explain It (B2)

First I sort the orders, then I return only the top five.

103.5 Memory Hook

Sort.

Then limit.

104. Practice Challenge

Return the top three customers alphabetically.

105. Question 10 — Remove Duplicate Cities

105.1 Recruiter Psychology

DISTINCT is simple but frequently tested.

105.2 Business Task

List every city only once.

105.3 SQL Solution

```
SELECT DISTINCT city  
FROM customers;
```

105.4 Explain It (B2)

DISTINCT removes duplicate values.

105.5 Memory Hook

DISTINCT means unique.

105.6 Common Mistake

Using GROUP BY unnecessarily.

106. Question 11 — Find Products in Electronics

106.1 Business Task

Show products from the Electronics category.

106.2 SQL Solution

```
SELECT product_name,  
       price  
FROM products  
WHERE category = 'Electronics';
```

106.3 Explain It (B2)

I filter products by category.

107. Question 12 — Sort Products by Price

107.1 Business Task

Show the cheapest products first.

107.2 SQL Solution

```
SELECT product_name,  
       price  
FROM products  
ORDER BY price ASC;
```

107.3 Explain It (B2)

I sort products from the lowest price to the highest.

108. SQL Interview Vocabulary

Word	Meaning
Filter	Remove unwanted rows
Sort	Change order
Return	Show results
Column	Vertical field
Row	Single record

Use these words during interviews.

109. Mini Mock Interview

109.1 Interviewer

Show all customers.

Pause.

Answer.

109.2 Interviewer

Show completed orders above £200.

Pause.

Answer.

109.3 Interviewer

Return the five largest orders.

Pause.

Answer.

110. Cheat Sheet

110.1 SELECT

```
SELECT column  
FROM table;
```

110.2 WHERE

```
WHERE condition;
```

110.3 ORDER BY

```
ORDER BY column DESC;
```

110.4 LIMIT

```
LIMIT 5;
```

110.5 DISTINCT

```
SELECT DISTINCT column;
```

111. Memory Hooks Summary

Concept	Hook
SELECT	Choose
WHERE	Filter rows
ORDER BY	Organise
LIMIT	Stop early
DISTINCT	Unique

112. End-of-Sprint Challenge

Without looking at previous examples, solve these tasks.

1. Show customers from London.
2. Show orders above £150.
3. Show completed orders.
4. Show the five largest orders.
5. List unique cities.
6. Show Electronics products sorted by price.

Target:

- under **8 minutes**
- explain every query aloud in English.

Congratulations.

You have completed **SQL Sprint 1**.

113. SQL Bible

114. Sprint 2 — Aggregations

114.1 GROUP BY • HAVING • Aggregate Functions

Congratulations.

You already know how to retrieve data.

Now you'll learn how analysts **summarise business performance**, which is exactly what hiring managers expect from a Junior Data Analyst.

115. Why Aggregations Matter

Most business questions are not:

Show me every order.

Instead they sound like:

- Which city generates the most revenue?
- How many customers signed up last month?
- What is the average order value?
- Which category sells best?

All of these require aggregation.

116. Memory Map

Think of the process like making piles.

Rows become groups.

Each group gets a calculation.

Example:

City	Orders
London	25
Manchester	18
Leeds	9

Rows disappear.

Groups remain.

117. Aggregate Functions Overview

Function	Purpose
COUNT()	Count rows
SUM()	Total value
AVG()	Average
MIN()	Smallest
MAX()	Largest

Memory Hook:

Count • Sum • Average • Minimum • Maximum

118. Question 13 — Count All Orders

118.1 Recruiter Psychology

Interviewers often begin with COUNT() because it's simple but essential.

118.2 Business Task

How many orders exist?

118.3 SQL Solution

```
SELECT COUNT(*) AS total_orders  
FROM orders;
```

118.4 Explain It (B2)

COUNT returns the number of rows.

118.5 Memory Hook

COUNT counts records.

118.6 Common Mistake

Using COUNT(amount) instead of COUNT(*).

119. Question 14 — Count London Customers

119.1 Business Task

Count customers living in London.

119.2 SQL Solution

```
SELECT COUNT(*) AS london_customers
FROM customers
WHERE city = 'London';
```

119.3 Explain It (B2)

First I filter London customers, then I count them.

120. Question 15 — Total Revenue

120.1 Recruiter Psychology

Revenue questions appear constantly.

120.2 Business Task

Calculate total sales.

120.3 SQL Solution

```
SELECT SUM(amount) AS total_revenue  
FROM orders;
```

120.4 Explain It (B2)

SUM adds every order amount together.

120.5 Memory Hook

SUM builds money.

121. Question 16 — Average Order Value

121.1 Business Task

Find the average order value.

121.2 SQL Solution

```
SELECT AVG(amount) AS average_order  
FROM orders;
```

121.3 Explain It (B2)

AVG calculates the typical order value.

121.4 Common Mistake

Confusing average with total.

122. Question 17 — Largest Order

122.1 Business Task

Find the highest order.

122.2 SQL Solution

```
SELECT MAX(amount) AS largest_order  
FROM orders;
```

122.3 Explain It (B2)

MAX returns the highest value.

123. Question 18 — Smallest Order

123.1 SQL Solution

```
SELECT MIN(amount) AS smallest_order  
FROM orders;
```

123.2 Explain It (B2)

MIN finds the lowest value.

124. Understanding GROUP BY

Imagine these rows.

Customer	Amount
Alice	120
Alice	80
Bob	60

GROUP BY creates:

Customer	Revenue
Alice	200
Bob	60

Rows become summaries.

Memory Hook:

GROUP BY creates piles.

125. Question 19 — Revenue by Customer

125.1 Recruiter Psychology

This is one of the most common interview questions.

125.2 Business Task

Calculate revenue for each customer.

125.3 SQL Solution

```
SELECT customer_id,  
       SUM(amount) AS revenue  
FROM orders  
GROUP BY customer_id;
```

125.4 Explain It (B2)

I group orders by customer and calculate total revenue.

125.5 Common Mistake

Forgetting GROUP BY.

126. Question 20 — Orders by Status

126.1 Business Task

Count completed and cancelled orders.

126.2 SQL Solution

```
SELECT status,  
       COUNT(*) AS orders  
FROM orders  
GROUP BY status;
```

126.3 Explain It (B2)

Every status becomes one group.

127. Question 21 — Customers by City

127.1 Business Task

Count customers in each city.

127.2 SQL Solution

```
SELECT city,  
       COUNT(*) AS customers  
FROM customers  
GROUP BY city;
```

127.3 Explain It (B2)

Each city becomes one group.

128. GROUP BY Rule

Every selected column must be either:

- grouped
- aggregated

Example.

Correct.

```
SELECT city,  
       COUNT(*)  
FROM customers  
GROUP BY city;
```

Wrong.

```
SELECT city,  
       first_name  
FROM customers  
GROUP BY city;
```

Memory Hook:

Every passenger needs a seat.

129. Question 22 — Average Revenue by City

129.1 Business Task

Calculate average order value by city.

129.2 SQL Solution

```
SELECT city,  
       AVG(amount) AS average_revenue  
FROM customers  
JOIN orders USING(customer_id)  
GROUP BY city;
```

129.3 Explain It (B2)

I join customers and orders, group by city and calculate the average.

Don't worry about JOIN syntax yet.

Sprint 3 explains it fully.

130. HAVING

WHERE filters rows.

HAVING filters groups.

Memory Hook.

WHERE before GROUP.

HAVING after GROUP.

131. Question 23 — Cities with More Than Five Customers

131.1 Recruiter Psychology

WHERE vs HAVING is a classic interview trap.

131.2 SQL Solution

```
SELECT city,  
       COUNT(*) AS customers  
FROM customers  
GROUP BY city  
HAVING COUNT(*) > 5;
```

131.3 Explain It (B2)

I first create city groups, then remove groups with five or fewer customers.

132. Question 24 — Customers Spending Over £500

132.1 Business Task

Find customers whose total spending exceeds £500.

132.2 SQL Solution

```
SELECT customer_id,  
       SUM(amount) AS revenue  
FROM orders  
GROUP BY customer_id  
HAVING SUM(amount) > 500;
```

132.3 Explain It (B2)

HAVING filters the grouped results.

133. WHERE vs HAVING

WHERE	HAVING
Filters rows	Filters groups
Before grouping	After grouping

Interview Tip.

If you're calculating something like SUM or COUNT, think HAVING.

134. Question 25 — Top Revenue Cities

134.1 Business Task

Show cities ranked by revenue.

134.2 SQL Solution

```
SELECT city,  
       SUM(amount) AS revenue  
FROM customers  
JOIN orders USING(customer_id)  
GROUP BY city  
ORDER BY revenue DESC;
```

134.3 Explain It (B2)

I calculate revenue for every city and sort from highest to lowest.

135. Business Case

Imagine you're interviewing at Tesco.
The hiring manager asks.

Which city performs best?

Your answer.

1. Join customers and orders.
2. Group by city.
3. Calculate revenue.
4. Sort descending.

Notice.

You're describing thinking.
Not just SQL.

136. Mini Mock Interview

136.1 Interviewer

How many customers do we have?

Pause.

Answer.

136.2 Interviewer

Which city has the highest revenue?

Pause.

Answer.

136.3 Interviewer

Which customers spent over £500?

Pause.

Answer.

137. Cheat Sheet

137.1 COUNT

```
COUNT(*)
```

137.2 SUM

```
SUM(amount)
```

137.3 AVG

```
AVG(amount)
```

137.4 GROUP BY

```
GROUP BY customer_id
```

137.5 HAVING

```
HAVING SUM(amount) > 500
```

138. Memory Hooks

Concept	Hook
COUNT	Count records
SUM	Build money
AVG	Typical value
GROUP BY	Make piles
HAVING	Filter groups

139. End-of-Sprint Challenge

Complete these without looking.

1. Count all customers.
2. Count London customers.
3. Calculate total revenue.
4. Find average order value.
5. Find the largest order.
6. Revenue by customer.
7. Customers by city.
8. Cities with more than five customers.
9. Customers spending over £500.
10. Top revenue cities.

Target.

- **12 minutes**
- explain every query aloud in English.

Congratulations.

You have completed **SQL Sprint 2**.

140. SQL Bible

141. Sprint 3 — JOINS, CTEs & Subqueries

141.1 The Language of Relationships

Welcome to the most important chapter for Junior Data Analyst interviews.

Most real business questions require combining information from multiple tables.

Instead of memorising JOIN syntax, you'll learn to think like an analyst.

142. The Evenec Retail Relationship Map

Our dataset follows one business story.

- Customers place Orders.
- Orders contain Items.
- Items reference Products.
- Orders receive Payments.

142.1 Entity Relationship Overview

customers

orders

payments

order_items

products

customer_id order_id order_id product_id Memory Hook:

Customer → Order → Item → Product

143. Why JOIN Exists

Imagine two separate tables.

customers

customer_id	first_name
1	Alice
2	Bob

orders

order_id	customer_id	amount
101	1	120
102	2	60

A JOIN connects them.

Result.

first_name	amount
Alice	120
Bob	60

144. JOIN Mental Model

Instead of remembering syntax,
remember relationships.

Keys connect stories.

Primary Key.
Foreign Key.
Bridge.

145. Question 26 — INNER JOIN

145.1 Recruiter Psychology

This is one of the most common interview questions.

145.2 Business Task

Show every order together with the customer's name.

145.3 SQL Solution

```
SELECT c.first_name,  
       o.order_id,  
       o.amount  
FROM customers c  
INNER JOIN orders o  
ON c.customer_id = o.customer_id;
```

145.4 Explain It (B2)

I connect customers and orders using customer_id and return customer names with their orders.

145.5 Memory Hook

INNER = Both must exist. _____

146. INNER JOIN Visual

Customers Orders Only the overlap remains.

147. Question 27 — LEFT JOIN

147.1 Business Task

Show every customer, even if they never ordered.

147.2 SQL

```
SELECT c.first_name,  
       o.order_id  
FROM customers c  
LEFT JOIN orders o  
ON c.customer_id = o.customer_id;
```

147.3 Explain It (B2)

Every customer appears. Missing orders become NULL.

147.4 Memory Hook

Left keeps everyone.

This phrase will appear throughout the book.

148. LEFT JOIN Visual

Customers Orders Everything from the left table survives.

149. Interview Trap

Question.

Why is LEFT JOIN useful?

Good answer.

It allows me to keep every customer even when related records don't exist.

Avoid saying.

Because it joins left.

Explain the business value.

150. Question 28 — Find Customers Without Orders

150.1 SQL

```
SELECT c.first_name  
FROM customers c  
LEFT JOIN orders o  
ON c.customer_id = o.customer_id  
WHERE o.order_id IS NULL;
```

150.2 Explain It (B2)

I keep every customer and filter those without matching orders.

Memory Hook.

NULL means “missing”.

151. Question 29 — Join Three Tables

Business questions often need multiple joins.

Example.

Customers.

Orders.

Payments.

```
SELECT c.first_name,  
       o.order_id,  
       p.payment_method  
FROM customers c  
JOIN orders o  
ON c.customer_id = o.customer_id  
JOIN payments p  
ON o.order_id = p.order_id;
```

Explain.

First I connect customers to orders, then orders to payments.

Think.

Step by step.

152. JOIN Order

Always ask yourself.

What is my starting table?

Then walk through relationships.

Customer

↓

Order

↓

Payment

Never jump randomly.

153. CTE — Common Table Expression

Interviewers love CTEs because they improve readability.
Memory Hook.

CTE = Temporary workspace.

Instead of writing one giant query,
build it in steps.

154. Question 30 — First CTE

Business Task.

Find high-value customers.

```
WITH customer_revenue AS (  
    SELECT customer_id,  
           SUM(amount) AS revenue  
    FROM orders  
    GROUP BY customer_id  
)  
SELECT *  
FROM customer_revenue  
WHERE revenue > 500;
```

154.1 Explain It (B2)

First I create a temporary table, then I filter it.

Memory Hook.

WITH creates a temporary workspace.

155. Why CTE Is Better

Instead of.

```
SELECT ...
```

inside another huge query,

split the problem.

Cleaner.

Easier to explain.

Interviewers appreciate readability.

156. Subqueries

A subquery is simply

a query inside another query.

Memory Hook.

Box inside a box.

157. Question 31 — Above Average Orders

Business Task.

Find orders larger than the average.

```
SELECT order_id,  
       amount  
FROM orders  
WHERE amount >  
(  
  SELECT AVG(amount)  
  FROM orders  
);
```

157.1 Explain It (B2)

First SQL calculates the average, then it compares every order against it.

158. Subquery Visual

Main Query

AVG(amount) One query feeds another.

159. Question 32 — Products Above Average Price

```
SELECT product_name,  
       price  
FROM products  
WHERE price >  
(  
    SELECT AVG(price)  
    FROM products  
);
```

Explain.

Products become filtered using the average price.

160. CTE vs Subquery

CTE	Subquery
Easier to read	Shorter
Reusable	Quick calculations
Better for interviews	Fine for simple tasks

Interview Tip.

If readability matters,
choose a CTE.

161. Business Case — Amazon Style

Imagine this question.

Find customers whose spending is above the company average.

Think aloud.

1. Calculate customer revenue.
2. Calculate average revenue.
3. Compare.

Interviewers score your reasoning,
not just the final SQL.

162. Mini Mock Interview

Interviewer.

Explain LEFT JOIN.

Pause.

Answer.

Interviewer.

Why use a CTE?

Pause.

Answer.

Interviewer.

Find orders above average.

Pause.

Answer.

163. Cheat Sheet

163.1 INNER JOIN

```
INNER JOIN table  
ON key = key;
```

163.2 LEFT JOIN

```
LEFT JOIN table  
ON key = key;
```

163.3 CTE

```
WITH temp AS (...)  
SELECT ...
```

163.4 Subquery

```
WHERE value >  
(  
SELECT AVG(...)  
)
```

164. Memory Hooks Summary

Concept	Hook
INNER	Both exist
LEFT	Left keeps everyone
NULL	Missing
CTE	Temporary workspace
Subquery	Box inside a box

165. End-of-Sprint Challenge

Complete these without looking.

1. Join customers and orders.
2. Show every customer using LEFT JOIN.
3. Find customers without orders.
4. Join customers, orders and payments.
5. Build a CTE.
6. Find orders above average.
7. Find products above average price.
8. Explain INNER JOIN in English.
9. Explain LEFT JOIN in English.
10. Explain why CTE improves readability.

Target.

- **15 minutes**
- explain every solution aloud in English.

Congratulations.

You have completed **SQL Sprint 3**.

166. SQL Bible

167. Sprint 4 — Window Functions & Interview Finale

167.1 ROW_NUMBER • RANK • DENSE_RANK • LAG • LEAD • OVER()

Welcome to the final SQL sprint.

By now you've learned:

- filtering;
- aggregation;
- joins;
- CTEs;
- subqueries.

Now you'll learn the functions that often separate a confident Junior Analyst from someone who only knows basic SQL.

168. Why Window Functions Matter

Imagine this question.

Who are the top three customers in every city?

GROUP BY cannot answer this alone.

Window functions calculate values **without destroying individual rows**.
Memory Hook.

GROUP BY collapses.

WINDOW looks through.

169. The Window Mental Model

Think of standing inside a moving window.

The table stays.

The calculation changes.

Rows remain visible.

170. OVER()

Every window function uses:

```
OVER( . . . )
```

Think of it as:

Look through this window.

171. Question 33 — ROW_NUMBER()

171.1 Recruiter Psychology

This is one of the most common technical interview questions.

171.2 Business Task

Number every order.

```
SELECT order_id,  
       amount,  
       ROW_NUMBER() OVER(ORDER BY amount DESC) AS row_num  
FROM orders;
```

171.3 Explain It (B2)

I sort orders by amount and assign a unique number to each row.

Memory Hook.

ROW_NUMBER never repeats.

172. Visual

Amount	ROW_NUMBER
500	1
400	2
300	3

Unique numbering.

173. Question 34 — RANK()

Business Task.

Rank customers by revenue.

```
SELECT customer_id,  
       revenue,  
       RANK() OVER(ORDER BY revenue DESC) AS customer_rank  
FROM customer_revenue;
```

Result.

Revenue	Rank
500	1
400	2
400	2
300	4

Notice.

Rank skips numbers.

Memory Hook.

RANK leaves gaps.

174. Question 35 — DENSE_RANK()

Same example.

```
SELECT customer_id,  
       revenue,  
       DENSE_RANK() OVER(ORDER BY revenue DESC) AS dense_rank  
FROM customer_revenue;
```

Result.

Revenue	Dense Rank
500	1
400	2
400	2
300	3

No gaps.

Memory Hook.

Dense fills gaps.

175. Interview Trap

Question.

When would you choose DENSE_RANK instead of RANK?

Good answer.

When tied values shouldn't create gaps in the ranking.

176. PARTITION BY

Window functions become much more powerful.

Example.

Rank customers **inside each city**.

```
SELECT customer_id,  
       city,  
       revenue,  
       ROW_NUMBER()  
       OVER(  
         PARTITION BY city  
         ORDER BY revenue DESC  
       ) AS city_rank  
FROM customer_revenue;
```

Memory Hook.

PARTITION creates mini competitions.

177. Visual

London.

Revenue	Rank
500	1
300	2

Manchester.

Revenue	Rank
600	1
250	2

Separate rankings.

178. Question 36 — Top Customer in Every City

Business Task.

Return only the best customer from each city.

```
WITH ranked AS
(
  SELECT customer_id,
         city,
         revenue,
         ROW_NUMBER()
           OVER(
             PARTITION BY city
             ORDER BY revenue DESC
           ) AS rn
  FROM customer_revenue
)

SELECT *
FROM ranked
WHERE rn=1;
```

Explain.

First I rank customers inside each city, then I keep only rank one.

179. LAG()

LAG compares with the previous row.

Business Task.

Compare today's revenue with yesterday's.

```
SELECT order_date,  
       revenue,  
       LAG(revenue)  
         OVER(ORDER BY order_date) AS previous_day  
FROM daily_sales;
```

Memory Hook.

LAG looks back.

180. LEAD()

LEAD looks forward.

```
SELECT order_date,  
       revenue,  
       LEAD(revenue)  
         OVER(ORDER BY order_date) AS next_day  
FROM daily_sales;
```

Memory Hook.

LEAD looks ahead.

181. Running Total

A classic interview question.

Business Task.

Calculate cumulative revenue.

```
SELECT order_date,  
       amount,  
       SUM(amount)  
       OVER(  
         ORDER BY order_date  
       ) AS running_total  
FROM orders;
```

Explain.

The total grows after every row.

Memory Hook.

Running total keeps growing.

182. Moving Average

Another popular interview task.

```
SELECT order_date,  
       revenue,  
       AVG(revenue)  
       OVER(  
         ORDER BY order_date  
         ROWS BETWEEN 6 PRECEDING  
         AND CURRENT ROW  
       ) AS seven_day_average  
FROM daily_sales;
```

Interview Tip.

Don't memorise.

Understand.

The window moves.

183. Window Function Cheat Sheet

Function	Purpose
ROW_NUMBER	Unique numbering
RANK	Ranking with gaps
DENSE_RANK	Ranking without gaps
LAG	Previous row
LEAD	Next row
SUM OVER	Running total

184. Business Case 1 — Tesco

Question.

Which city generated the most revenue?

Expected Thinking.

- Join customers.
 - Group by city.
 - Sum revenue.
 - Sort descending.
-

185. Business Case 2 — Amazon

Question.

Find the highest-spending customer in every city.

Expected Solution.

Use:

- CTE
 - ROW_NUMBER
 - PARTITION BY
-

186. Business Case 3 — Deliveroo

Question.

Find customers who haven't ordered recently.

Expected Thinking.

- Latest order.
 - Compare dates.
 - Filter inactive customers.
-

187. Business Case 4 — NHS

Question.

Calculate average appointments per patient.

Same SQL principle.

Business changes.

Logic stays.

188. Business Case 5 — Wise

Question.

Which payment method processes the most money?

Expected Thinking.

- Join payments.
 - Group by payment method.
 - SUM(amount).
-

189. Business Case 6 — Monzo

Question.

Find unusually large transactions.

Expected Thinking.

Use:

- AVG
 - Subquery
 - WHERE.
-

190. Business Case 7 — Retail Dashboard

Question.

Create KPIs.

Required.

- Total revenue
 - Average order
 - Largest order
 - Customer count
-

191. Business Case 8 — Customer Retention

Question.

Who made repeat purchases?

Expected Thinking.

Count orders.

HAVING COUNT()>1. _____

192. Business Case 9 — Product Performance

Question.

Top-selling category.

Expected Thinking.

Join.

Group.

Sort.

193. Business Case 10 — Executive Report

Imagine presenting to a CEO.
Don't say.

GROUP BY...

Instead say.

London generated the highest revenue, while Birmingham showed the fastest growth.

Analysts communicate business impact.

194. Full Mock SQL Interview (45 Minutes)

This simulates a real UK Junior Data Analyst interview.

194.1 Part 1 — Warm-up (10 min)

Interviewer

Show London customers.

Pause.

Answer.

Find completed orders.

Pause.

Answer.

Top five orders.

Pause.

Answer.

194.2 Part 2 — Aggregations (10 min)

Revenue by city.

Pause.

Answer.

Customers spending over £500.

Pause.

Answer.

Explain HAVING.

Pause.

Answer.

194.3 Part 3 — JOINS (10 min)

Connect customers and orders.

Pause.

Answer.

Find customers without orders.

Pause.

Answer.

Explain LEFT JOIN.

Pause.

Answer.

194.4 Part 4 — Advanced SQL (15 min)

Rank customers.

Pause.

Answer.

Above-average orders.

Pause.

Answer.

Top customer in every city.

Pause.

Answer.

Running total.

Pause.

Answer.

195. Interview English Cheat Sheet

Useful phrases.

Instead of:

“SQL does this...”

Say:

First I filter...

Then I group...

After that I calculate...

Finally I sort...

This sounds much more natural in interviews.

196. Common Interview Mistakes

✗ Writing before thinking.
✓ Explain your approach first.

✗ Memorising syntax only.
✓ Explain business reasoning.

✗ Panicking during joins.
✓ Draw relationships mentally.

197. 5-Minute Emergency Revision

Remember these.

197.1 **SELECT**

Choose columns.

197.2 **WHERE**

Filter rows.

197.3 **GROUP BY**

Create groups.

197.4 **HAVING**

Filter groups.

197.5 **JOIN**

Connect stories.

197.6 **CTE**

Temporary workspace.

197.7 **ROW_NUMBER**

Unique numbering.

197.8 **RANK**

Leaves gaps.

197.9 **DENSE_RANK**

No gaps.

197.10 **LAG**

Looks back.

197.11 **LEAD**

Looks ahead.

198. Final SQL Challenge

Complete these ten tasks without looking.

1. London customers.
2. Revenue by city.
3. Top five orders.
4. Customers over £500.
5. Join customers and orders.
6. Customers without orders.
7. Above-average orders.
8. Top customer per city.
9. Running total.
10. Rank customers.

Target.

- **25 minutes**
 - explain every query aloud in English.
-

199. SQL Bible Complete

Congratulations.

You have completed all four SQL sprints.

Before moving to Excel and Power BI, make sure you can:

- write queries without copying;
- explain every query in English;
- recognise business questions behind SQL tasks.

Remember.

Companies don't hire analysts because they know SQL.

They hire analysts because they solve business problems with SQL.

200.

201.